

Neural-network-based molecular dynamics simulations reveal that proton transport in water is doubly gated by sequential hydrogen-bond exchange

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The transport of excess protons in water is central to acid–base chemistry, biochemistry and energy production. However, elucidating its mechanism has been challenging. Recent nonlinear vibrational spectroscopy experiments could not be explained by existing models. Here we use both vibrational spectroscopy calculations and neural-network-based molecular dynamics simulations that account for nuclear quantum effects for all atoms to determine the proton transport mechanism. Our simulations reveal an equilibrium between two stable proton-localized structures with distinct Eigen-like and Zundel-like hydrogen-bond motifs. Proton transport follows a three-step mechanism gated by two successive hydrogen-bond exchanges: the first reduces the proton-acceptor water coordination, leading to proton transfer, and the second, the rate-limiting step, prevents rapid back-transfer by increasing the proton-donor coordination. This sequential mechanism is consistent with experimental characterizations of proton diffusion, explaining the low activation energy and the prolonged intermediate lifetimes in vibrational spectroscopy. These results are crucial for understanding proton dynamics in biochemical and technological systems.

Proton transport in water governs aqueous acid–base reaction kinetics, powers cellular metabolism and controls energy production in fuel cells. However, despite decades of intensive experimental and theoretical investigations, its mechanism still has not been fully elucidated. Proton transport is generally understood to occur through a proton-transfer relay between hydrogen (H)-bonded water molecules, which results in its anomalously rapid diffusion compared with other ions. However, no consensus has been reached on the molecular mechanism of proton transport, the nature of its rate-limiting step and even the equilibrium structure of an excess proton in water.

Two limiting structures have been proposed for the hydrated excess proton—the Eigen complex¹, where a central hydronium ion is symmetrically solvated by three water molecules as $\text{H}_3\text{O}^+(\text{H}_2\text{O})_3$, and the

Zundel complex^{2,3} H_5O_2^+ , where the proton is equally shared between two flanking water molecules—and the proton is usually considered to adopt a broad distribution of structures between these limits. A major computational effort^{4–12} begun three decades ago has concluded that the Eigen form is more stable and that proton transport goes through an unstable Zundel transition state^{6–9,12,13}. However, recent time-resolved vibrational spectroscopy results^{14–17} suggest that distorted Zundel-like structures are long-lived, contradicting this picture. This contradiction compounds another unresolved question about the nature of the rate-limiting step: the most recent proposed mechanisms^{8,9,13} involve the concerted rearrangement of at least two H-bonds, a requirement inconsistent with the measured very low diffusion activation energy¹⁸. Altogether, these results therefore call for a revised picture of proton transport.

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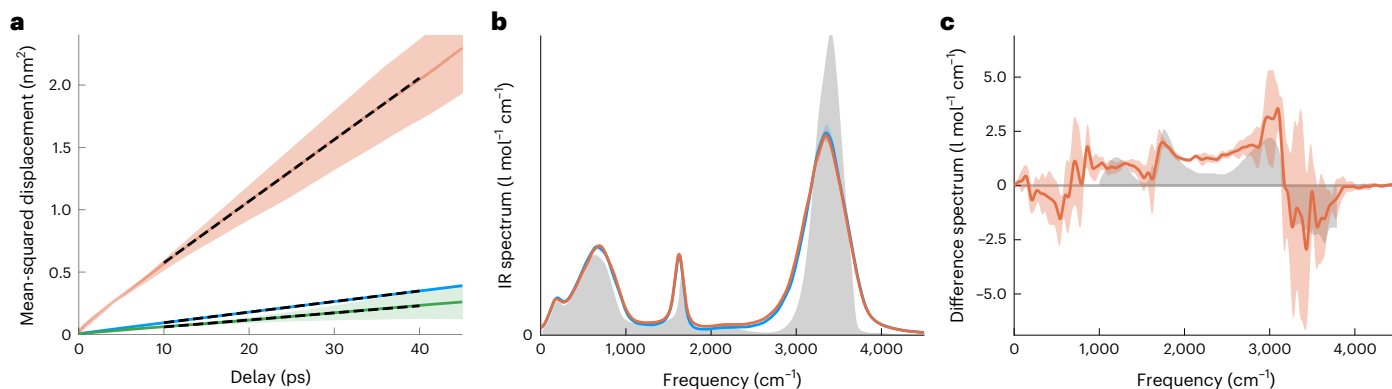


Fig. 1 | Diffusion constant and vibrational spectra. **a**, Mean-squared displacements of the excess proton (orange), of the excess proton undergoing vehicular diffusion in the absence of proton transfers (green) and of water molecules (blue) calculated from 20 independent 200 ps trajectories for the free simulations propagated with TRPMD using a DNN trained at the revPBE0-D3 level (revPBE0-D3/TRPMD is shown in ref. 48 to provide an excellent description of neat water structure, dynamics and vibrational spectra) and from 5 independent 200 ps trajectories for the constrained simulations (vehicular diffusion, green). Diffusion coefficients are determined from the slope of the mean-squared displacement on the 10–40 ps interval (dashes) and corrected for finite-size effects (Supplementary Fig. 9). The shaded regions show the 95% confidence

interval derived from 20 independent trajectories (excess proton diffusion, orange; water diffusion, blue) or 5 independent trajectories (excess proton vehicular diffusion, green). **b**, Calculated IR spectra of the acidic (orange) and neat water (blue) solutions, with the experimental spectrum (grey) of liquid water (from ref. 49); the spectrum is calculated as the Fourier transform of the dipole time-derivative autocorrelation function, with the molecular dipoles obtained from separately trained DNNs (Methods). The shaded regions show the 95% confidence interval derived from five independent trajectories. **c**, Difference IR spectrum between acidic and neat water solutions from our simulations (orange) and from experiments¹⁶ (grey). The shaded regions show the 95% confidence interval derived from five independent trajectories.

To address this question, we have performed and analysed extensive neural network potential (NNP)-based molecular dynamics (MD) simulations combined with vibrational spectra calculations. Simulating proton transport requires combining an accurate description of electronic structure rearrangements caused by covalent bond breaking and making, an explicit description of nuclear quantum effects associated with high-frequency hydrogen motions^{6,19,20}, and trajectories long enough to resolve subtle stability differences between proton structures. Because of the large computational costs involved, previous simulations with ab initio and empirical valence bond MD had to compromise on some of these points. However, recent developments in machine learning now offer a solution to all three requirements via deep neural network (DNN) potentials that reproduce high-level reference electronic structure calculations at a fraction of the cost, allowing for unprecedented trajectory lengths²¹. We have trained a DNN potential at the revPBE0-D3 hybrid density functional theory (DFT) level and used it to perform simulations of an excess proton in water at ambient conditions using thermostatted ring-polymer molecular dynamics (TRPMD)²² that explicitly account for nuclear quantum effects of all atoms in the system.

Results and discussion

Diffusion coefficient and linear vibrational spectroscopy

We first confirm that this technique quantitatively captures the hydrated proton's anomalously fast diffusion and vibrational spectral features (Fig. 1). Our simulations yield accurate descriptions both for the excess proton diffusion coefficient ($8.57 \pm 0.05 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$, in very good agreement with the $9.3 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$ experimental value²³) and the ratio between water and proton diffusion coefficients ($D_{\text{H}_2\text{O}}/D_{\text{H}^+} = 0.207 \pm 0.002$ from our simulations and 0.25 experimentally²³). The key role played by actual proton transfers in this enhanced diffusion is evidenced by the much-reduced vehicular diffusion coefficient ($D_{\text{vehicular}} = 1.32 \pm 0.01 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$) of a hydronium ion whose internal covalent bonds are maintained intact. In addition, we verify that our computed infrared (IR) spectrum of an excess proton in water accurately reproduces all the vibrational hallmarks of the hydrated proton reported experimentally^{14–17,24} (Fig. 1b,c and Methods).

Proton transfer mechanism

We then determine the molecular mechanism of individual proton transfer events from our simulations. Characterizing this mechanism is particularly challenging due to the ultrafast dynamics involved. The excess proton undergoes sustained transfers⁶ every 100 fs or less based on changes in the defect-carrying oxygen atom identity observed in simulations^{8,25} and on the ultrafast proton structural dynamics measured by nonlinear vibrational spectroscopy^{15,26}. In contrast, successful proton hops leading to diffusion measured by nuclear magnetic resonance²⁷ and two-dimensional IR spectroscopy²⁸ take place on a ten-times-slower picosecond timescale. This discrepancy is known²⁹ to arise from the multiple back and forth transfers between the same pair of water molecules before a stable hop occurs and leads to diffusion. However, distinguishing this transient rattling from stable transfer events and determining what controls the slower hops is a major difficulty for simulations. Filtering procedures have been proposed^{15,9,30,31}, but they all require arbitrary selections that typically ignore all transfers back to the previous proton-carrying oxygen, an exclusion contrary to the uniform random walk that predicts that the latter should receive one-third of the transfers.

To overcome these difficulties and focus on the intrinsic rearrangements of the proton environment, we consider two key coordinates. The first one is the energy gap ΔE between two reference configurations where the proton is moved from the donor to the acceptor. This reports on the proton potential asymmetry and on whether it favours the proton on the donor ($\Delta E < 0$) or acceptor ($\Delta E > 0$) side (Fig. 2a). It is analogous to the electron transfer theory solvent coordinate and has already been successfully applied to proton transfer^{19,32}. Its key advantage relative to the extensively used^{6,9,12,33} proton position coordinate is that large but transient OH bond elongations are not spuriously considered as transfers when the proton potential asymmetry keeps the proton localized on the donor. Out of the three hydrogen atoms in the hydronium ion, we focus here on the best candidate for proton transfer, that is, the one with the least asymmetric proton potential. A second coordinate is required to monitor H-bond rearrangements around the donor and the acceptor, since bulk water molecules form approximately four H-bonds, donating two and accepting two, while the hydronium ion is coordinated by only three H-bond acceptors.

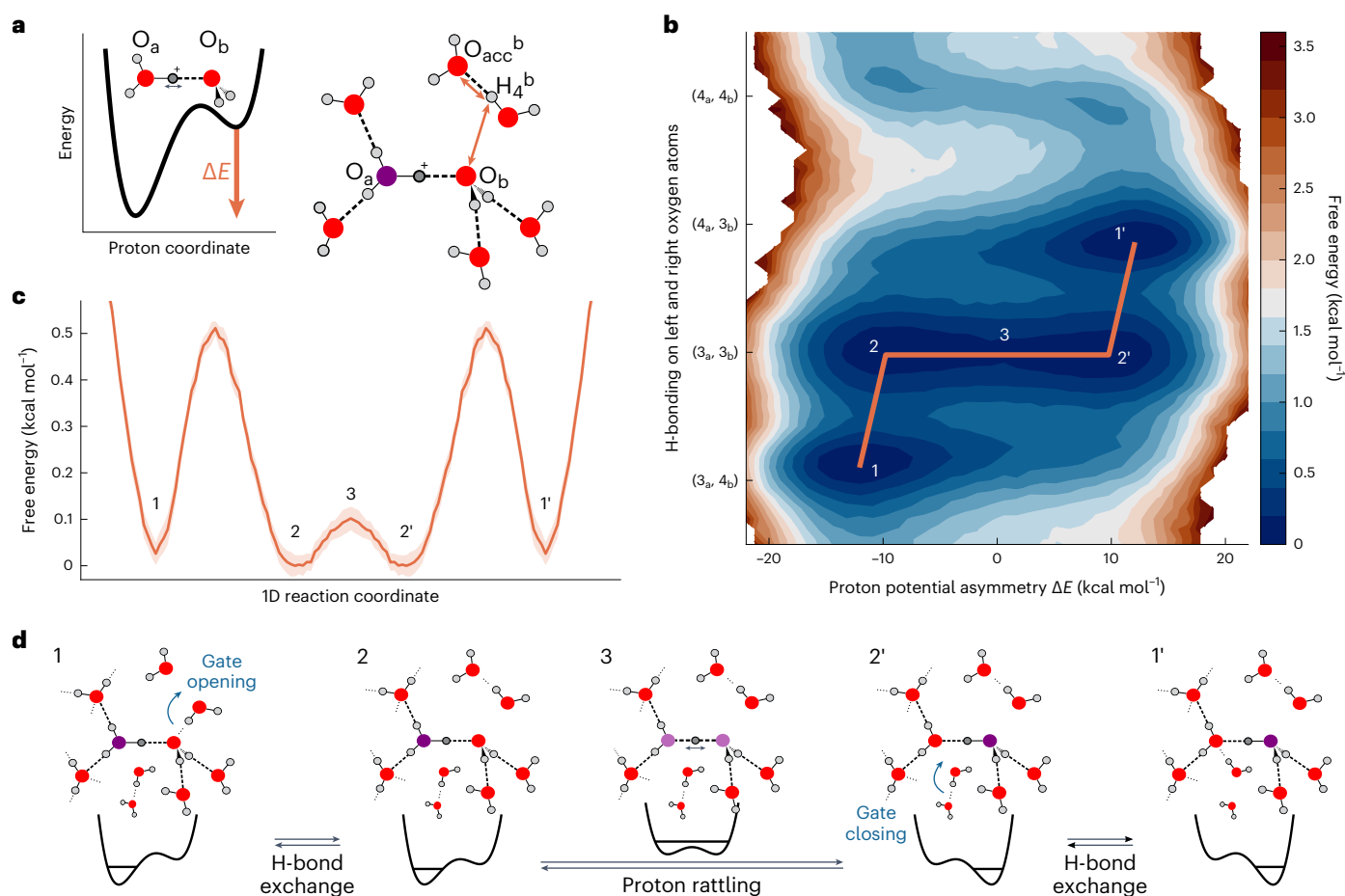


Fig. 2 | Proton transfer mechanism. **a**, Definition of coordinates: proton potential asymmetry ΔE (left) and periodic H-bond coordinate probing the donor and acceptor numbers of H-bonded partners (right). **b**, Free energy surface along the proton potential asymmetry ΔE and H-bond coordinate noted (n_a, n_b) where $n_{a,b}$ is the number of H-bond partners for $O_{a,b}$, focusing on the best O_b proton acceptor candidate with the smallest $|\Delta E|$ asymmetry. The uncertainty on the free energy is <0.07 kcal mol⁻¹ (Supplementary Fig. 2 and Supplementary Table 1).

c, Free energy profile along the pathway shown in **b**. Shaded regions show the 95% confidence interval derived from 20 independent trajectories. **d**, Schematic proton transfer mechanism with typical H-bond arrangements and proton potential energy profiles for key locations on the surface, where the transferred proton and defect-carrying oxygen atom are highlighted. While the Eigen-like (1 and 1') and Zundel-like (2 and 2') structures are stable, the ideal Zundel structure (3) is unstable.

H-bond rearrangements were suggested early on to be key for proton transfer¹⁹, and simulation studies have all stressed their importance^{4–6,8,9,11–13,34}. However, distinguishing between transient and stable H-bond reorganizations is critical. In water, H-bond breaks are frequent but transient because dangling H-bond configurations are unstable³⁵. Stable H-bond rearrangements occur only when the H-bond donor jumps to another available H-bond partner³⁶. This provides a natural definition for stable H-bond configurations. For the proton donor and acceptor oxygens, we examine whether the best fourth H-bond partner is closer to this oxygen or to another H-bond acceptor. This results in a continuous and periodic coordinate that monitors whether the proton-donor and proton-acceptor oxygen atoms have three or four stable H-bonded neighbours (Fig. 2a, Methods and Extended Data Fig. 1).

The free energy surface calculated along these two coordinates from our 4-ns-long unconstrained MD simulations (Fig. 2b) reveals critical features of the proton structure and transfer mechanism not visible in previous analyses based on other more transient fluctuation-prone coordinates.

Concerning the equilibrium proton structure, it has been assumed in the literature that only one of the Eigen and Zundel forms is stable. Consequently, the focus has been on determining which structure is more stable, treating the other as the transition state crossed during

proton transfer. In contrast, the free energy surface in Fig. 2b reveals an equilibrium between two distinct stable H-bond arrangements while the proton remains bonded to the same water molecule. When the proton is favoured on the donor oxygen O_a ($\Delta E < 0$), O_a is three-coordinate as expected around a hydronium ion (Supplementary Fig. 5), but the acceptor oxygen O_b exhibits two stable H-bond arrangements, three- and four-coordinate, that we denote $(3_a, 3_b)$ and $(3_a, 4_b)$, respectively. Two additional free energy basins are found for the equivalent structures with the proton on O_b when $\Delta E > 0$. The $(3_a, 4_b)$ and $(4_a, 3_b)$ structures resemble the Eigen complex on O_a and O_b , respectively, where the hydronium ion is solvated by three bulk-like four-coordinated water molecules. In contrast, in the $(3_a, 3_b)$ structure, O_a and O_b share a symmetric solvation environment where both have lost an H-bond: the H-bond arrangement is that expected around the Zundel form, and we will refer to it as ‘Zundel-like’. However, the covalent bonding structure is not the idealized Zundel complex since the strong proton potential asymmetry $|\Delta E|$ keeps the proton localized on O_a or O_b (Supplementary Fig. 7). The distances between the hydronium ion and its hydration shell water molecules exhibit a hierarchy of H-bond lengths in quantitative agreement with recent X-ray results on protonated water complexes³⁷; these distances do not change between the Eigen-like and Zundel-like states, but their gradation is markedly enhanced in the Zundel transition state (Supplementary Fig. 6).

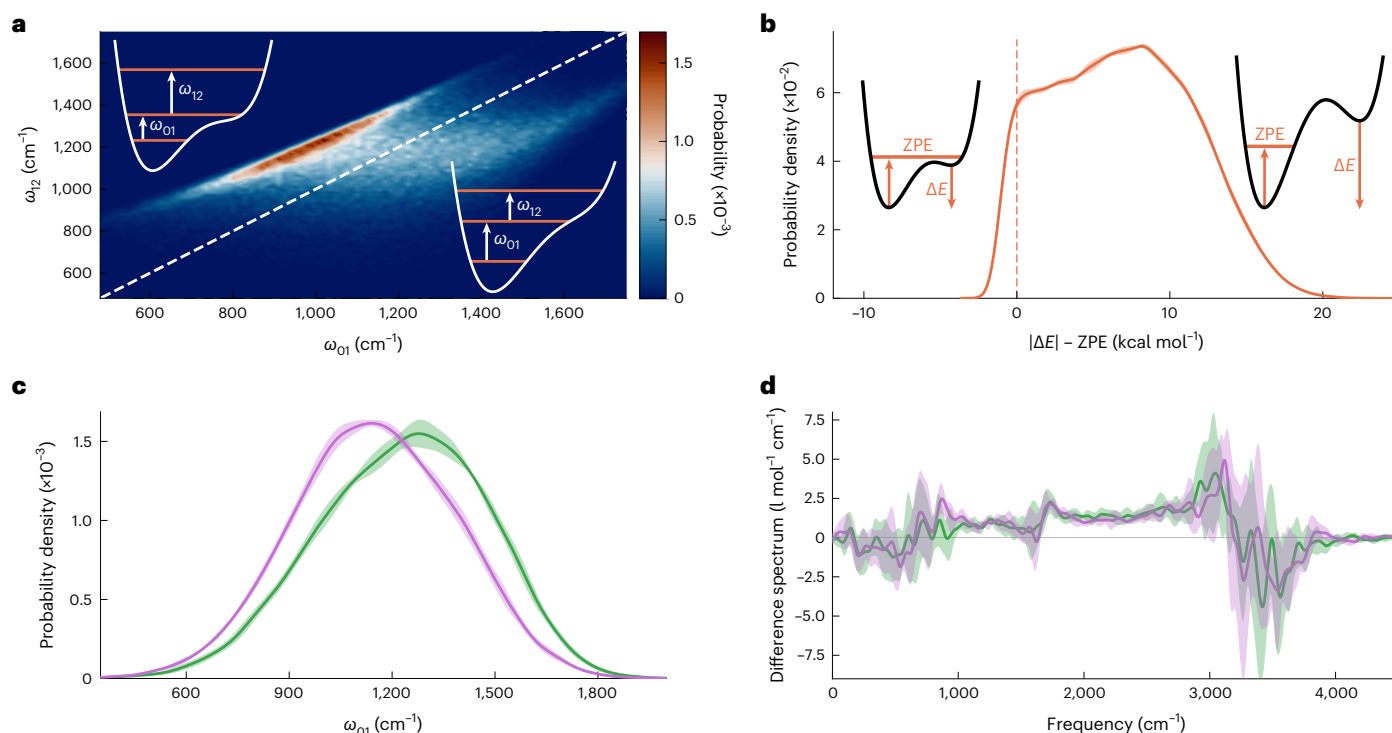


Fig. 3 | Vibrational fingerprints of H-bond configurations. **a**, Correlation diagram between the $\hbar\omega_{01}$ ground-state and $\hbar\omega_{12}$ excited-state absorption energies, determined from a discrete-variable representation calculation of the vibrational quantum energy levels for the proton-transfer mode (Methods and Supplementary Section 3). **b**, Probability distribution of the difference between proton transfer mode zero-point energy ZPE and proton potential asymmetry $|\Delta E|$ (see inset), showing that the proton is most frequently localized ($|\Delta E| - \text{ZPE} > 0$). Shaded regions show the 95% confidence interval derived from

20 independent trajectories. **c**, Probability distributions of the ω_{01} frequency in the Eigen-like (green) and Zundel-like (purple) H-bond arrangements. The shaded regions show the 95% confidence interval derived from 20 independent trajectories. **d**, Difference spectra between acidic solution and neat water decomposed into Eigen-like (green) and Zundel-like (purple) H-bond arrangements. The shaded regions show the 95% confidence interval derived from five independent trajectories.

Our simulations thus show that the hydrated excess proton equilibrium structure is characterized by two features: (1) its covalent bonding pattern, which is Eigen-like and keeps the proton localized on a single water molecule, and (2) its H-bond arrangement in equilibrium between Eigen-like and Zundel-like stable solvation structures in almost equal proportions (this equilibrium is affected by, for example, temperature and isotopic substitution; Extended Data Figs. 2 and 3).

This previously unnoticed structural equilibrium carries major implications for proton transfer. The most probable transfer mechanism is determined from the minimum free energy path for inverting the proton potential asymmetry from $\Delta E < 0$ to $\Delta E > 0$ on the surface (Fig. 2b,c); the associated transfer rates between intermediates are obtained from a time-correlation function analysis of our simulations. As seen in Fig. 2b,d, systems starting from the Eigen-like H-bond configuration (state 1) first require the breaking of an H-bond on the acceptor to reach the Zundel-like H-bonded state (state 2). This occurs on a 2.1 ± 0.4 ps timescale on each acceptor, slightly faster than typical H-bond exchanges in bulk water³⁶ and, thus, three times faster on a 0.7 ± 0.2 ps timescale when the H-bond break can occur on any of the three acceptors. Once in the symmetric H-bond arrangement where both the donor and acceptor have a hydronium-like coordination, facile proton transfer can occur, overcoming a low and mostly entropic barrier (Extended Data Fig. 3) on a 0.6 ± 0.1 ps timescale, and taking the system through the ideal fully symmetric $\Delta E = 0$ Zundel structure (state 3). This results in a Zundel-like H-bond configuration where the proton is now on the acceptor side (state 2').

Our proposed mechanism shares some features with previous theoretical works but differs in several critical points that are essential

for explaining the experimental observations. First, we emphasize that our multistep mechanism with two stable configurations contrasts with all previously proposed mechanisms that involve a single stable Eigen state and an unstable Zundel transition state. The initial step in our mechanism involves the breaking of the H-bond on the acceptor, in agreement with early pioneering studies on proton transfer^{4,6,8,19,38} that had recognized its essential role. However, a key distinction arises as our results indicate that this step is not rate-limiting, in contrast to the conclusion of these early studies. Subsequent simulations^{9,11,13,29,30} suggested that additional H-bond rearrangements are involved, but the exact nature of the limiting step remained elusive. A presolvation picture was proposed^{9,39}, involving the same H-bond breaking on the acceptor and H-bond formation on the donor as in our mechanism. However, in this model, both H-bond rearrangements must occur concertedly before the proton is transferred^{9,39}. Such a concerted mechanism would be expected to have a proton diffusion activation energy typically twice that of water self-diffusion, since the latter involves a single H-bond exchange⁴⁰. However, experiments indicate that proton transport has a smaller activation energy than water diffusion¹⁸. As we will show, this discrepancy is solved by our sequential mechanism, which involves the successive breaking of one H-bond, proton transfer and the reformation of another H-bond. Finally, the proton donor–acceptor pair in our symmetrically under-coordinated H-bond arrangement shares similarities with the previously proposed special pair^{4,8}, including its H-bond configuration and strong H-bond interaction, but our simulations do not show the suggested⁸ rapid <100 fs switching of the partner's identity among the central hydronium H-bond acceptors, since this requires stable

H-bond breaking and making on the pair acceptor that occur on a picosecond timescale.

Nonlinear vibrational spectroscopy

A major challenge to the previously proposed mechanisms has recently come from ultrafast vibrational spectroscopy experiments^{14–17} that conclude that Zundel-like structures persist for several hundred femtoseconds and, thus, cannot be a transition state. This is based on three key observations. First, the $1,200\text{ cm}^{-1}$ proton transfer mode exhibits a positive anharmonicity, that is, the vibrationally excited-state absorption energy $\hbar\omega_{12}$ is larger than the ground-state one $\hbar\omega_{01}$, which indicates a broad proton well with steep walls as found in the Zundel complex but not in typical Eigen structures^{15,16}. Two additional results—the slow recovery of the proton transfer mode frequency distribution after excitation¹⁵ and the long-lived coupling between the redshifted OH stretch and blueshifted bend modes¹⁶—reveal a long-lived structural heterogeneity and suggest that the Zundel-like state is stable.

We now demonstrate that our simulations and mechanism explain these observations. We show that the species identified by ultrafast vibrational spectroscopy is strongly evocative of the Zundel-like H-bond arrangement revealed in our simulations and that the long lifetimes observed by spectroscopy are caused by the distinct vibrational properties of the Eigen-like and Zundel-like stable H-bond configurations.

We determine ω_{01} and ω_{12} from vibrational quantum energy level calculations for the proton-transfer mode (Methods). The correlation between the calculated ω_{01} and ω_{12} in Fig. 3a exhibits a maximum probability for $\omega_{12}/\omega_{01} \approx 1.21 \pm 0.01$, in excellent agreement with the 1.1 experimental anharmonicity¹⁶. However, while previous studies had proposed that such ω_{12}/ω_{01} values would be characteristic of a delocalized (symmetric¹⁵ or asymmetric¹⁶) Zundel structure, our results show that the proton potential asymmetry $|\Delta E|$ is larger than the proton zero-point energy and, thus, keeps the proton localized on the donor (Fig. 3b).

The proton-transfer ω_{01} frequency distribution redshifts by $\sim 100\text{ cm}^{-1}$ between the Eigen-like and Zundel-like H-bond configurations (Fig. 3c) because of the stronger O_2H H-bond, a feature also reflected in the shorter donor–acceptor distance (Extended Data Fig. 4b). These different spectral densities imply that the proton transfer mode after excitation recovers its equilibrium frequency distribution on the timescale of the H-bond rearrangement on the proton acceptor; this is $\sim 0.7\text{ ps}$ in our calculations, which explains the long-lived inhomogeneity observed experimentally¹⁵ (our interpretation is further supported by the absence of such inhomogeneity in experiments on H_5O_2^+ in acetonitrile¹⁵, consistent with the lack of H-bond donors in this aprotic solvent). We further note that the observed proton-transfer mode's strong IR and weak Raman activities³³ are explained by the H-bonded pair undercoordination-induced symmetry breaking around the proton-carrying oxygen atom, without requiring the proton to be delocalized. For frequencies above the proton-transfer mode, the calculated difference spectra in Fig. 3d show that both H-bond arrangements display redshifted OH stretch and blueshifted bend modes, a result consistent with the experimental observation of a single spectroscopic species in this frequency range¹⁶ and with previous calculations^{25,33}. Although establishing an exact connection between a vibrational spectrum and a molecular structure remains challenging, complicated by the different possible choices of coordinates to describe the variety of proton structures, our results reveal an equilibrium between two stable structures, with different H-bond arrangements and different spectral densities for the proton transfer mode. Thus, our simulations provide a molecular interpretation for the proton structural dynamics observed in ultrafast vibrational spectroscopy, which could not be explained by previous models with a single stable configuration.

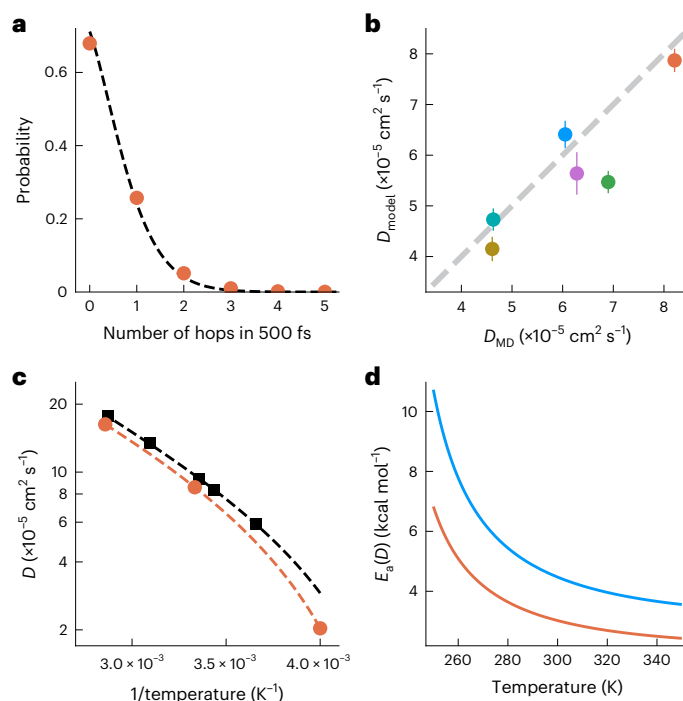


Fig. 4 | Proton transport. **a**, Probability distribution of the number n of stable hops during a $T = 500\text{ fs}$ interval and Poisson distribution $(1/n!)(T/\tau)^n \exp(-T/\tau)$ expected from the hopping time $\tau = 1.47\text{ ps}$ (dashes). The error bars representing the 95% confidence interval derived from 20 independent trajectories are smaller than the orange dots. **b**, Correlation between the finite-size proton diffusion coefficients obtained from our simulations D_{MD} and from our model combining random walk due to proton hops and vehicular diffusion D_{model} (Methods). In addition to the revPBE0-D3 TRPMD simulations in H_2O described in the text (orange), we have repeated our calculations with heavy water D_2O at the same level (green), without the TRPMD description of nuclear quantum effects (classical, blue) and with a DNN trained on B3LYP-D3 electronic structure calculations (quantum H_2O , purple; quantum D_2O , gold; classical H_2O , turquoise). The error bars show the 95% confidence interval derived from 20 independent trajectories. **c**, Temperature dependence of the excess proton diffusion coefficient from our revPBE0-D3 TRPMD simulations (orange dots) and experiments⁴³ (black squares), with power-law fits (dashes; Supplementary Section 5.3). The error bars on simulated diffusion coefficients representing the 95% confidence interval derived from 20 independent trajectories are smaller than the orange dots. **d**, Diffusion activation energy of the excess proton (orange) and of water (blue) from power-law fits.

Proton transport

We now determine how individual proton transfer events are related to long-range proton transport through the liquid, identify the diffusion rate-determining step and relate its activation energy to experimental measurements. Our analysis of the free energy surface in Fig. 2b shows that H-bond breaking on the acceptor is limiting only for the fraction of systems with an Eigen-like H-bond arrangement. Once the Zundel-like H-bond arrangement is obtained, the proton transfer can proceed easily and fast proton rattling via forward and backward transfers between the same donor–acceptor pair occurs as long as this H-bond arrangement persists. This implies that the limiting step for stable proton hopping on the acceptor is the H-bond reformation on the donor to prevent any back transfer. Proton transport is thus controlled not only by stable H-bond breaking on the acceptor that opens the gate in front of the proton and allows the proton transfer, but also mostly by stable H-bond reformation on the donor that subsequently closes the gate behind the proton, preventing fast back-transfer and leading to a doubly gated ‘pull–push’ mechanism (Fig. 2d). These two events are sequential and not concerted, leading to the intermittent

rattling phases observed in the simulations. Our calculated hopping time is 1.47 ± 0.05 ps (Methods and Extended Data Fig. 4a), in very good agreement with the experimental 1.6–1.7 ps timescale^{27,28}. In addition, our identification of the rate-determining step as the H-bond formation on the donor results in a low barrier aligning with the recent experimental estimation of <1 kcal mol⁻¹ for the proton transport free energy barrier⁴¹. It also explains the experimentally measured acceleration of proton diffusion by weak H-bond accepting anions⁴¹ that facilitate the H-bond exchange from the anion to the donor, whereas this could not be accounted for by other mechanisms that exclusively involve H-bond cleavage at oxygen sites as rate-determining steps.

It has been suggested^{10,30,31} that multiple concerted hops could transfer the proton over several molecules, leading to ‘burst’ phases with much faster proton diffusion than during ‘rest’ phases. Our results do not support this suggestion. Our analysis of stable proton hops using the above donor H-bond reformation criteria shows that the probability of multiple jumps follows the Poisson distribution expected for a series of independent events (Fig. 4a). In agreement with previous work⁴², this shows that the correlated transfers found in some studies arise from transient rattling but disappear when only stable hops are considered.

This supports a proton transport picture involving independent single-proton hops, and we model the diffusion constant as a random walk whose waiting time is the delay between successive stable hops and whose amplitude is the oxygen–oxygen distance, added to the vehicular contribution coming from the hydronium-ion diffusion in the absence of proton transfer. Figure 4b shows that this model quantitatively reproduces the simulated proton diffusion coefficient and captures the impacts of isotopic substitution, nuclear quantum effects and changes in the electronic structure level.

We have repeated our TRPMD simulations at 250 K and 350 K. Our simulated proton and water diffusion coefficients quantitatively reproduce the experimental temperature dependence (Fig. 4c), including the lower activation energy for proton diffusion than for water diffusion (Fig. 4d). This temperature study adds further support to our identification of the H-bond formation on the donor as the rate-limiting step, since the enthalpic barrier of this step determined from the enthalpic contribution to the two-dimensional free energy surface (Extended Data Fig. 5) is $\Delta H^\ddagger \approx 2.0$ kcal mol⁻¹, yielding the activation energy $E_a \approx \Delta H^\ddagger + k_B T \approx 2.6$ kcal mol⁻¹. This is in good agreement with the values from our simulations (3.0 kcal mol⁻¹) and from experiments (2.8 kcal mol⁻¹ (ref. 43)) and smaller than the 4.5 kcal mol⁻¹ water diffusion activation energy⁴⁰, as observed experimentally. Our proposed mechanism thus explains the low activation energy of proton diffusion, and as shown in Extended Data Fig. 3, its sequential nature involving successive barriers with different temperature responses leads to interesting experimentally accessible temperature (and isotope) effects, including in the supercooled regime.

Conclusion

The proton transport mechanism determined from our simulations shows that proton diffusion in water is a multistep process, limited by two successive stable H-bond exchanges, the key one being the reformation of a H-bond on the former proton donor after proton transfer. This sequential mechanism identifies the long-lived structures detected by recent experiments and explains the low diffusion activation energy. H-bond jump exchanges have already been shown to govern water dynamics and have been extensively characterized^{36,40,44}. This insight will be crucial to determine how proton transport is affected in acid–base aqueous solutions⁴⁵, in situations with different H-bonding environments including clusters²⁴, protonated water complexes^{15,17,37} and aqueous interfaces, and in biochemical and technological systems whose properties are controlled by proton diffusion, including, for example, concentrated solutions and solutions in confinement^{46,47}.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41557-024-01593-y>.

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Methods

NNPs

NNPs are trained with DeePMD-kit 2.0 (ref. 50). The embedding network yielding the descriptors is made of three layers of 20, 40 and 80 neurons, and the fitting network giving the atomic energies and forces is made of four layers of 240 neurons. The embedding is performed according to the smooth edition scheme⁵¹ with a 6 Å cut-off. Atomic NNPs are constructed for each chemical element in the system, here O and H. Each NNP is trained for 1,000,000 batches by minimizing a loss function applied to the mean squared errors on the sum of atomic energies and on the forces predicted by the network compared with reference calculations on the same configurations. This minimization is performed with an exponential learning rate that decays from 10^{-3} at the beginning of the procedure to a final value of 10^{-8} with a decay rate of 0.944. Adjustable prefactors for the energy and force error terms are used, which put an emphasis on forces for the first batches (weights of 0.01 and 1,000 for the energies and forces, respectively), and equal weights for forces and energies are used in the final batches (1 for both).

Hybrid functional DFT with dispersion correction was shown to provide a good description of liquid water. We selected the B3LYP-D3 functional owing to its wide use in chemistry and the revPBE0-D3 functional that was recently shown to yield an excellent description of the structure, dynamics and vibrational spectroscopy of neat liquid water when combined with nuclear quantum effects⁴⁸. All reference electronic structure calculations are performed with the CP2K software with Goedecker-Teter-Hutter (GTH) pseudopotentials for core electrons and triple- ζ (TZV2P) basis set for valence electrons, with the auxiliary density matrix method. The target accuracy for self-consistent field convergence is set to 3×10^{-8} .

The training system is a cubic box of 12.42 Å side with 64 water molecules and 1 excess proton. The starting point of our revPBE0-D3 and B3LYP-D3 NNPs is a 20-ps-long *ab initio* MD trajectory propagated at the B3LYP-D3 level, from which 1,000 configurations are evenly selected. The training set is iteratively augmented in five iterations using a query by committee approach with successive exploration, labelling and training steps following the method described in DP-GEN⁵². For this concurrent learning process, three NNPs are trained at each iteration and used to propagate a 20-ps-long six-bead PIGLET⁵³ trajectory with i-PI⁵⁴. The first set of configurations is removed from the training set after the first iteration. The model deviation between the three NNPs is computed from the bead configurations. Configurations with a maximal deviation on forces above 0.1 eV Å^{-1} are added to the training set. The iterative process is stopped when less than 7% of the configurations meet this criterion and less than 0.7% have a maximal deviation on forces above 0.15 eV Å^{-1} . At this stage, the dataset contains 2,188 B3LYP-D3 configurations. Energies and forces are calculated for these configurations at the revPBE0-D3 level to create the training set⁵⁵. Exploration with MD trajectories using the resulting NNP shows that the revPBE0-D3 NNP is converged and that no further iteration and addition to the training set is required (that is, the fraction of configurations requiring additional training is below the above criteria).

We confirmed that the NNPs trained on smaller (64 H_2O , H^+) systems provide a quantitative description of the forces for the larger (192 H_2O , H^+) system used for our TRPMD simulations.

The correlation between forces from reference DFT calculations and from our NNP evaluated on 40 uncorrelated configurations that were not included in the training set and were sampled every 100 ps from MD simulations is shown in Supplementary Fig. 1a,b. The NNP error remains below 0.1 eV Å^{-1} , and the ratio of the root mean squared error to the standard deviation is smaller than 2.5% for the revPBE0-D3 NNP and smaller than 5.5% for the B3LYP-D3 NNP. This shows that the NNPs provide reactive force-field descriptions of hybrid DFT quality.

MD simulations

Nuclear quantum effects are explicitly described for all atoms in our simulations via thermostatted ring-polymer molecular dynamics

(TRPMD)²². Simulations are performed with the i-PI software⁵⁴ using 32 beads per atom. Our simulated systems contain 1 excess proton in a box of 192 water molecules at the experimental water density. For each DFT level (revPBE0-D3 and B3LYP-D3) and each water isotope (H_2O and D_2O), 20 independent 200-ps-long TRPMD trajectories are propagated at 300 K with a 0.25 fs propagation timestep and configurations are saved every 5 fs. Five complementary 200-ps-long TRPMD simulations of H^+ in H_2O at the revPBE0-D3 level with a 0.5 fs output interval are performed for the IR spectroscopy calculations. Additional simulations using smaller boxes containing one excess proton and 64 water molecules are performed with the revPBE0-D3 NNP and TRPMD to determine finite-size corrections to the diffusion coefficient (Supplementary Fig. 9), and simulations where the hydronium internal covalent bonds are maintained intact are propagated for all NNPs and isotopes to determine the vehicular diffusion coefficient.

To assess the impact of nuclear quantum effects on proton transport, we further perform MD simulations using LAMMPS with the same NNPs but with classical nuclei. We propagate one trajectory of 4 ns with each kind of NNP (revPBE0-D3 or B3LYP-D3) and the same system as in the TRPMD simulations. A 0.5 fs propagation timestep, a Bussi–Parrinello velocity rescaling thermostat with a time constant of 100 fs and an output interval of 5.0 fs are used for these simulations. Both trajectories are split in 20 blocks of 200 ps for analysis.

Observables are computed independently for each trajectory of 200 ps (TRPMD simulations) or each block of 200 ps (classical nuclei simulations), and the values given in this paper are the average over all trajectories with an uncertainty corresponding to the standard deviation times Student coefficient for 95% confidence interval divided by the squared root of the number of trajectories.

Calculation of IR spectra and proton transfer mode energy levels

IR spectrum. The simulation box total dipole is determined a posteriori for each frame along the MD trajectory using the deep Wannier framework⁵⁶. A deep Wannier neural network (DWNN) is trained to determine the relative positions of Wannier centroids with respect to the closest oxygen atom, using the Wannier centres computed with CP2K for each configuration in our training set. The correlations for test set configurations between the Wannier centroid positions from the DFT calculation and those from the DWNN show that Wannier centroids are accurately predicted by the DWNN (Supplementary Fig. 1c,d). The DWNNs are used to compute the Wannier centroids and the total dipole vector in each trajectory frame with a 0.5 fs output interval. Following refs. 22,57, the IR spectra are calculated from the Fourier transform of the TRPMD approximation of the Kubo-transformed total dipole time derivative autocorrelation function

$$I(\omega) \propto \frac{1}{6cV\epsilon_0 k_B T} \int_{-\infty}^{+\infty} d\tau e^{i\omega\tau} \langle \dot{\mathbf{M}}(0) \cdot \dot{\mathbf{M}}(\tau) \rangle \quad (1)$$

where c is the speed of light, V the volume of the simulation box, ϵ_0 the vacuum permittivity, k_B the Boltzmann constant, T the temperature and $\mathbf{M}$ the total dipole.

Proton transfer mode energy levels. The proton transfer mode energy levels are obtained a posteriori in each trajectory frame by quantizing the transferred proton vibration while keeping all other atoms fixed at their centroid positions. The proton one-dimensional potential energy profile is calculated with the NNP and placing the proton on the $\text{O}_a\text{--O}_b$ donor–acceptor axis and scanning its position on a uniform grid with 100 points extending from 5% to 95% of the O–O distance (Supplementary Fig. 3). The quantum vibrational energy levels for the proton transfer mode are calculated by solving the vibrational Schrödinger equation using a discrete variable representation with a sinc-function basis set⁵⁸ and the iterative Lanczos algorithm, with a 50 kcal mol^{-1}

energy cut-off and a 0.03 Å constant spacing grid along the δ_{H} proton position coordinate (Supplementary Fig. 4).

Coordinates for the proton-transfer free energy surface

Proton potential asymmetry ΔE coordinate. The proton potential asymmetry is probed by the ΔE environment coordinate¹⁹ that reports on the potential energy difference between two reference proton locations while all other atoms in the system are kept fixed. It is defined as

$$\Delta E(\mathbf{R}) = V(\mathbf{r}_{\text{H}}^{\text{a}}; \mathbf{R}) - V(\mathbf{r}_{\text{H}}^{\text{b}}; \mathbf{R}), \quad (2)$$

where V is the NNP potential energy, and $\mathbf{R}$ is the TRPMD centroid positions (that is, the centres of mass of the ring polymer) of all atoms except the transferred proton. $\mathbf{r}_{\text{H}}^{\text{a,b}}$ is the reference H positions of the transferred proton defined as

$$\mathbf{r}_{\text{H}}^{\text{a,b}} = \frac{\mathbf{r}_{\text{a}} + \mathbf{r}_{\text{b}}}{2} \pm \delta_{\text{H}} \frac{\mathbf{r}_{\text{b}} - \mathbf{r}_{\text{a}}}{|\mathbf{r}_{\text{b}} - \mathbf{r}_{\text{a}}|}, \quad (3)$$

where $\mathbf{r}_{\text{a,b}}$ is the donor and acceptor $\text{O}_{\text{a,b}}$ oxygen atom positions, respectively, and δ_{H} is the distance between the proton and the middle of the $\text{O}_{\text{a,b}}$ flanking oxygen atom pair in the reference geometry, with $\delta_{\text{H}} = 0.3$ Å. This value approximately corresponds to the position of the potential energy profile minimum in the Eigen configuration (Supplementary Fig. 3). ΔE is computed a posteriori in each MD frame for all three hydrogen atoms on the defect-carrying oxygen atom. These hydrogen atoms are then rank-ordered to identify the proton with the smallest $|\Delta E|$ asymmetry, which is the one most likely to be transferred.

Hydrogen-bond configuration coordinate. We define a coordinate that probes the presence of stable fourth water H-bond donors on the proton donor O_{a} and acceptor O_{b} oxygen atoms. The following procedure is used to identify the best potential fourth H-bond donor around $\text{O}_{\text{a,b}}$ and to calculate the H-bond coordinate:

1. If the fourth-closest oxygen atom to $\text{O}_{\text{a,b}}$ donates an H-bond such that the $\text{HO}_{\text{a,b}}$ distance is less than 2.5 Å, this oxygen atom is selected as the fourth H-bond donor $\text{O}_4^{\text{a,b}}$. Otherwise, $\text{O}_4^{\text{a,b}}$ is the nearest oxygen neighbour within a 60° semiangle cone whose axis passes through $\text{O}_{\text{a,b}}$, is orthogonal to the plane containing the transferred proton and the two other hydrogen atoms covalently bonded to $\text{O}_{\text{a,b}}$ and is oriented away from the plane defined by the proton and hydrogen atoms (Extended Data Fig. 1). The hydrogen atom bonded to $\text{O}_4^{\text{a,b}}$ that is closest to $\text{O}_{\text{a,b}}$ is denoted $\text{H}_4^{\text{a,b}}$.
2. We identify the oxygen atom $\text{O}_{\text{acc}}^{\text{a,b}}$ that is the closest H-bond acceptor for $\text{H}_4^{\text{a,b}}$ distinct from $\text{O}_{\text{a,b}}$. We then determine whether $\text{O}_4^{\text{a,b}} - \text{H}_4^{\text{a,b}}$ donates a stronger H-bond to $\text{O}_{\text{a,b}}$ or to $\text{O}_{\text{acc}}^{\text{a,b}}$ by considering the difference

$$\delta_{\text{HB}}^{\text{a,b}} = \text{H}_4^{\text{a,b}} \text{O}_{\text{a,b}} - \text{H}_4^{\text{a,b}} \text{O}_{\text{acc}}^{\text{a,b}}. \quad (4)$$

3. The H-bond arrangement around the $\text{O}_{\text{a}} - \text{O}_{\text{b}}$ pair is described by the γ_{HB} H-bond coordinate defined by

$$\gamma_{\text{HB}}(\text{O}_{\text{a}} - \text{O}_{\text{b}}) = \arctan\left(\frac{\delta_{\text{HB}}^{\text{b}}}{\delta_{\text{HB}}^{\text{a}}}\right) + \pi[1 - \Theta(\delta_{\text{HB}}^{\text{a}})], \quad (5)$$

where Θ is the Heaviside function. The γ_{HB} coordinate is continuous and 2π -periodic and has up to four basins centred on $-\pi/4$, $\pi/4$, $3\pi/4$ and $5\pi/4$. These basins respectively correspond to $(3_{\text{a}}, 4_{\text{b}})$, $(3_{\text{a}}, 3_{\text{b}})$, $(4_{\text{a}}, 3_{\text{b}})$ and $(4_{\text{a}}, 4_{\text{b}})$ states, where the first and second numbers are the numbers of H-bonds formed by O_{a} and O_{b} .

H-bond hopping times

Exchange times between free energy basins are computed from side-side time-correlation functions

$$C_{ij}(t) = \langle p_i(0) p_j(t) \rangle, \quad (6)$$

where $p_i(t)$ is 1 when the system is in state i and 0 otherwise, with absorbing boundary conditions in the product state. We consider eight distinct states defined by ranges of γ_{HB} and ΔE coordinates centred on their local free energy minima. Their respective definitions are (1) $-1.178 < \gamma_{\text{HB}} < 0.393$, $-15 < \Delta E < -9$ kcal mol⁻¹, (2) $0.393 < \gamma_{\text{HB}} < 1.178$, $-13 < \Delta E < -2$ kcal mol⁻¹, (3) $1.1963 < \gamma_{\text{HB}} < 2.749$, $-13 < \Delta E < -2$ kcal mol⁻¹, (4) $3.534 < \gamma_{\text{HB}} < 4.320$, $-15 < \Delta E < -9$ kcal mol⁻¹ and the symmetric states with positive ΔE ranges for the four remaining states. The exchange time τ_{ij} from state i to state j is then determined as

$$\tau_{ij} = \frac{1}{C_{ij}(t \rightarrow \infty)} \int_0^\infty dt \left[1 - \sum_{k=1, k \neq i}^8 C_{ik}(t) \right]. \quad (7)$$

The proton stable hopping time is determined as the time to go from an initial three-coordinate defect-carrying oxygen state to a state where the same oxygen atom is four-coordinate. We therefore calculate the time-correlation function $C_{ij}(t)$ equation (6), where state i is defined by $-20 < \delta_{\text{HB}}^{\text{a}} < -0.7$ Å and state j is defined by $0.7 < \delta_{\text{HB}}^{\text{a}} < 20$ Å, with absorbing boundary conditions in state j (Extended Data Fig. 4a). The proton stable hopping time is then defined as

$$\tau = \int_0^\infty dt [1 - C_{ij}(t)]. \quad (8)$$

Random walk model for proton diffusion coefficient

The proton diffusion coefficient is modelled as resulting from a random walk due to proton hops and from vehicular diffusion,

$$D_{\text{model}} = \langle \text{OO}^2 \rangle / 6\tau + D_{\text{vehicular}}, \quad (9)$$

where $\langle \text{OO}^2 \rangle$ is the mean-squared distance between hydronium oxygen and H-bond acceptor oxygen atoms (Extended Data Fig. 4c and Supplementary Table 6), τ is the stable hopping time determined from a time-correlation function analysis (Supplementary Table 5) and $D_{\text{vehicular}}$ is the vehicular diffusion constant (Supplementary Table 3).

Data availability

Molecular configurations, energies and forces used for NNP training and the relative displacement of the Wannier centroids for DNN are publicly available on Zenodo at <https://doi.org/10.5281/zenodo.11965260> (ref. 55). Source data are provided with this paper.

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Author contributions

W.H.T. and D.L. conceptualized the project. A.G. performed the simulations. A.G., W.H.T. and D.L. analysed the data and wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

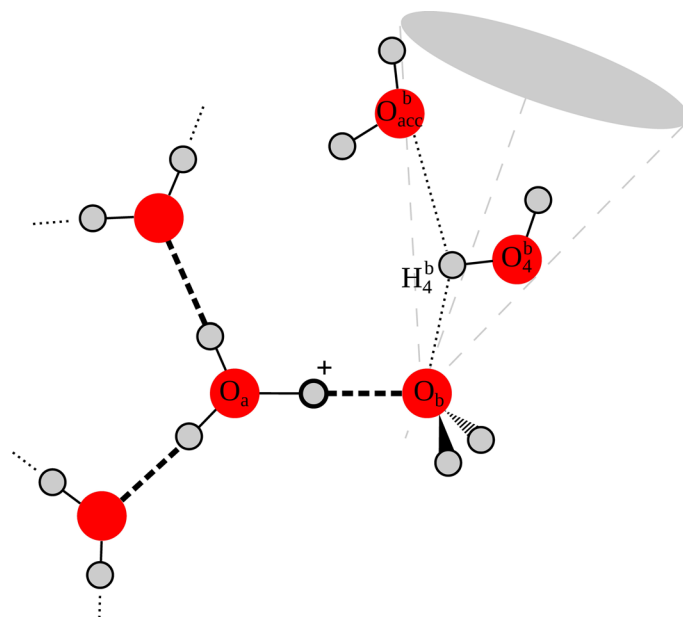
Extended data is available for this paper at <https://doi.org/10.1038/s41557-024-01593-y>.

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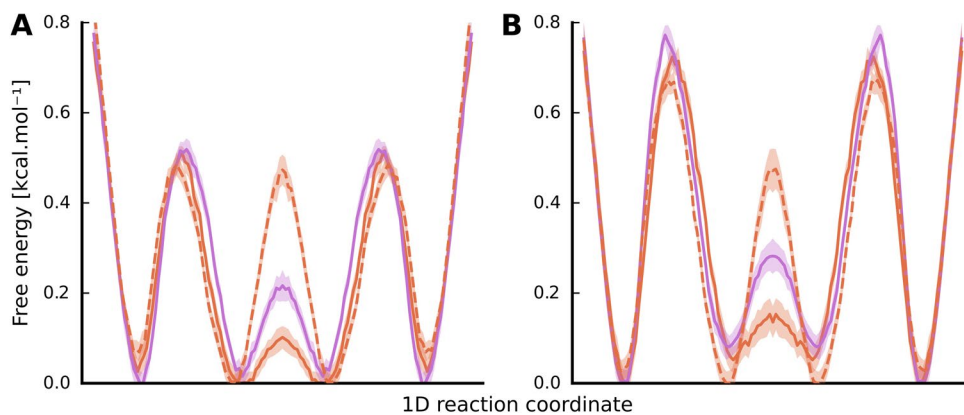
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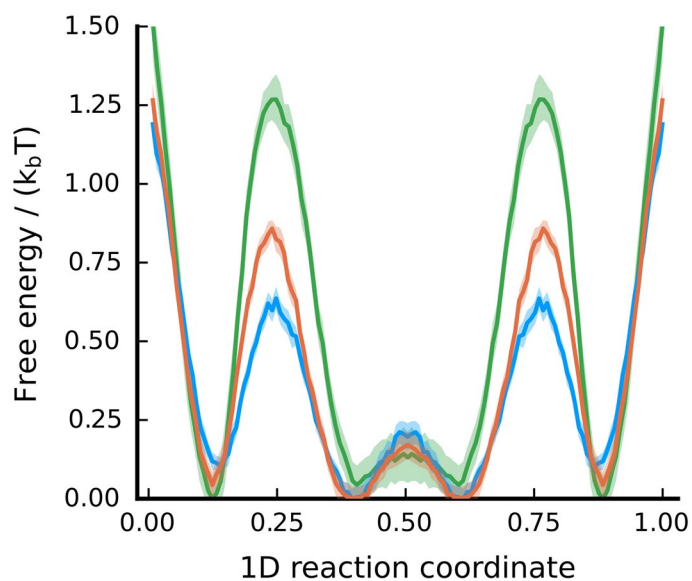


Extended Data Fig. 1 | Hydrogen-bond coordinate. Schematic representation of the atoms involved in the calculation of the H-bond coordinate around O_b .



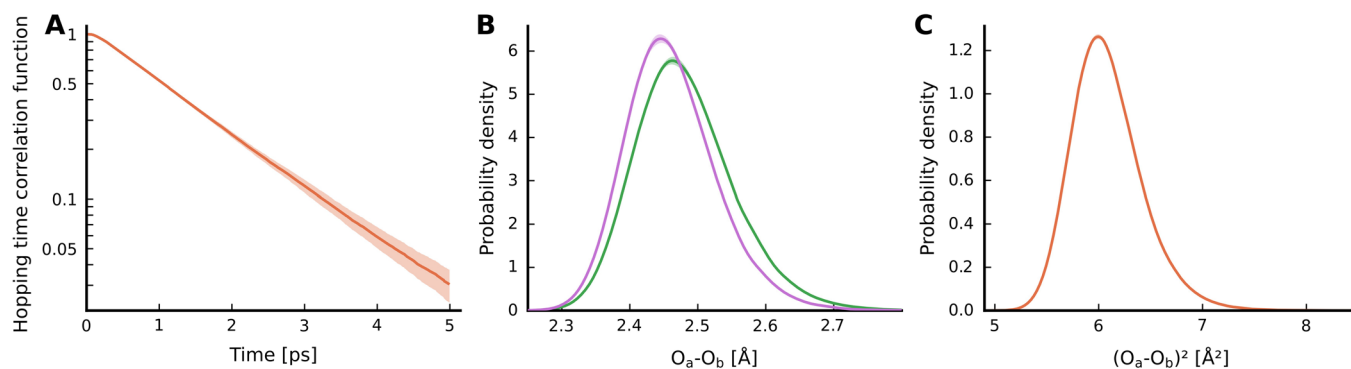
Extended Data Fig. 2 | Proton transfer free energy profiles in H₂O and in D₂O at 300 K. One-dimensional free energy profile along the reaction coordinate determined from the two-dimensional surface in Fig. 2b and Supplementary Fig. 2 (A) with the revPBE0-D3 NNP and (B) with the B3LYP-D3 NNP, respectively

for H₂O with nuclear quantum effects (solid orange lines), D₂O with nuclear quantum effects (solid purple lines) and H₂O with classical nuclei (orange dashes). Shaded regions show the 95% confidence interval derived from 20 independent trajectories.



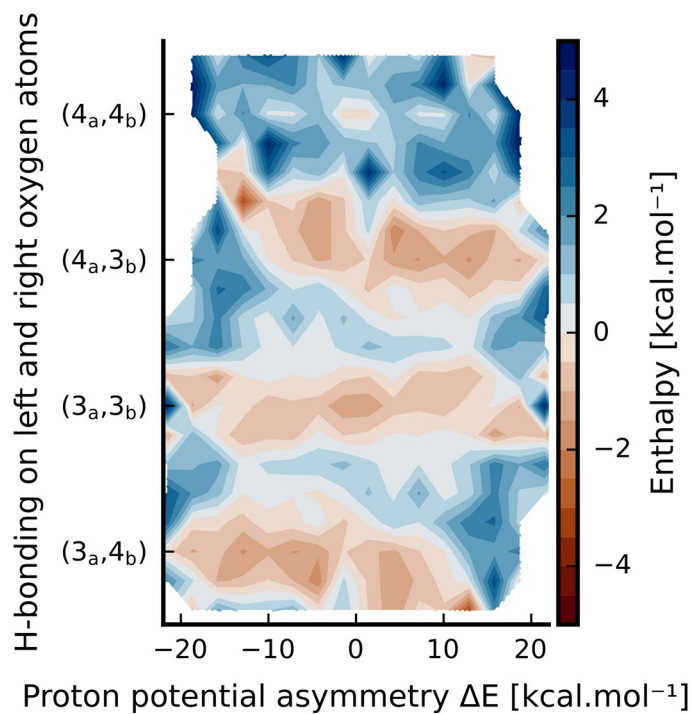
Extended Data Fig. 3 | Proton transfer free energy profiles normalized by the thermal energy. One-dimensional free energy profile along the reaction coordinate calculated with the revPBE0-D3 NNP in simulations of H₂O with nuclear quantum effects at 300 K (orange), 250 K (green) and 350 K (blue).

The very small temperature dependence of the central barrier in $k_B T$ units suggests that it is mostly of entropic origin. Shaded regions show the 95% confidence interval derived from 20 independent trajectories.



Extended Data Fig. 4 | Random walk modeling of the diffusion coefficient. (A) Stable proton hopping time-correlation function $1 - C_{ij}(t) = 1 - \langle p_i(0)p_j(t) \rangle$ (see definition in ref. 59) from our revPBE0-D3/TRPMD H^+ , H_2O simulation at 300 K. Shaded regions show the 95% confidence interval derived from 20 independent trajectories. (B) Distribution of O_a-O_b distances in Eigen-like ($3_a, 4_b$) (green) and Zundel-like ($3_a, 3_b$) (purple) H-bond arrangements from our revPBE0-D3/TRPMD

H^+ , H_2O simulation at 300 K. Shaded regions show the 95% confidence interval derived from 20 independent trajectories. (C) Distribution of $(O_a-O_b)^2$ squared distances in our revPBE0-D3/TRPMD H^+ , H_2O simulation at 300 K. Shaded regions show the 95% confidence interval derived from 20 independent trajectories.



Extended Data Fig. 5 | Proton transfer enthalpy surface. Enthalpic contribution to the proton transfer free energy surface in Fig. 2b computed at the revPBE0-D3 level from TRPMD simulations of H₂O at 300 K using fluctuation theory⁶⁰.